

SOVA KUSHWAHA

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PROFESSIONAL SUMMARY

Applied AI researcher focused on computer vision and deep learning, with professional software engineering experience and a practice-led research portfolio in Python. Skilled in model implementation, experimental evaluation, dataset organisation and clear reporting of technical results for academic and project stakeholders. Author of a peer-reviewed conference paper accepted for oral presentation at CISCom 2026 and for publication in Springer CCIS (Scopus) on explainable CNN classification of ovarian cysts in ultrasound imaging. Seeking the Aston University Research Assistant role to contribute to reproducible evaluation of AI systems for public-space video analytics.

TECHNICAL SKILLS

Research & AI: Machine Learning, Deep Learning (CNNs, Autoencoders), Computer Vision, Video Anomaly Detection, Model Evaluation (AUC, AP, operating points), False-Alert / False-Positive Analysis, Threshold Sensitivity, Trustworthy AI, Explainable AI, Statistical Testing

Programming & Tools: Python, PyTorch, OpenCV, scikit-learn, NumPy, pandas, SciPy, matplotlib, Jupyter, Git / GitHub, reproducible experiment logging

Data & Communication: Dataset preparation and documentation, quantitative result interpretation, technical reporting, figures/tables for research communication, stakeholder-facing explanation of model behaviour

Engineering: Structured debugging, reusable modules, version control, experiment records, clear written and oral technical communication

RESEARCH & PUBLICATIONS

Conference Paper (Accepted for Oral Presentation)

2025 / 2026

An Explainable VGG6 Convolutional Neural Network for Multi-Class Ovarian Cyst Classification in Ultrasound Imaging

- Accepted for oral presentation at the 2nd International Conference on Computational Intelligence and Soft Computing (CISCom 2026), Malta.
- Accepted for publication in Springer Communications in Computer and Information Science (CCIS) (Scopus-indexed).
- Designed and evaluated an explainable VGG6 CNN for multi-class ovarian cyst classification from ultrasound images, combining deep learning with explainable AI.
- Demonstrates end-to-end research practice: model design, evaluation, interpretability and academic communication.

RELEVANT RESEARCH PROJECT

False-Alert & Practical Reliability Evaluation (Public-Space Video Analytics)

2026

github.com/sovakushwaha/avenue-false-alert-reliability

- Built a reproducible PyTorch pipeline aligned with practical-reliability evaluation for public-space video anomaly detection using the CUHK Avenue benchmark (official frames and avenue.mat labels).
- Implemented and evaluated a Conv Autoencoder baseline for frame-level anomaly scoring; reported frame AUC 0.75 and AP 0.47, plus deployment-oriented metrics beyond AUC.
- Analysed false alerts, threshold sensitivity, event-level alert coalescing and estimated reviewer burden (minutes per hour) across operating points.
- Showed that event coalescing reduced apparent false alerts by about 20× versus frame-level rates at best F1 (about 19.9k to about 974 false alerts/hour under stated assumptions), while reviewer load can still exceed practical capacity.
- Documented recurring failure modes (FP bursts), produced report-ready figures and machine-readable metrics, and maintained open experiment records and code for reproducibility.

PROFESSIONAL EXPERIENCE

Flutter Developer

Mar 2025 to Oct 2025

Thulo Lalitpur, Nepal

- Delivered cross-platform product features from requirements through implementation and debugging.
- Practised structured problem solving, reusable code and clear technical communication in a delivery-focused team.
- Built habits of documentation and deadline-driven delivery transferable to research task planning with a PI.

Flutter Developer

Mar 2024 to Mar 2025

Gaaubesi Logistics Kathmandu, Nepal

- Contributed to mobile application development in an operational business setting.
- Worked with application data flows and product requirements to ship usable features.
- Collaborated across mixed-skill teams and maintained professional coding standards.

Digital Skills Tutor and Workshop Facilitator
Freelance Kathmandu and UK

- Designed and delivered structured digital-skills sessions for mixed-ability learners.
- Produced clear written guides and adapted explanations for non-specialist audiences.
- Used feedback to refine delivery, useful for research dissemination and team updates.

2023 to 2025

Tutor and Creator, SovaVerse (YouTube)
Independent educational channel youtube.com/@sovaVerse01

- Creates tutorials on Data Science, AI and Machine Learning for a broad audience.
- Explains technical concepts in accessible language for technical and non-technical viewers.
- Manages end-to-end content from research and scripting through publishing and iteration.

2023 to Present

SELECTED ADDITIONAL PROJECTS

SparkyBot, AI Voice Assistant

- Award-winning assistant project (voice, multimodal features). **2nd Prize**, KIST Inter-College IT Project Competition; presented to an external judging panel.

KIST IT Fair

Sales Insight Workflow (Python, SQL, SQLite)
github.com/sovakushwaha/sales-insight-workflow

- Built an end-to-end Python workflow across 4,200 validated sales line items, including source-quality checks, relational joins, feature engineering, KPI reporting and SQLite outputs.
- Organised datasets, ran reproducible analyses and interpreted quantitative results into clear business recommendations.

2026

Ecommerce Transactions EDA (Python)
github.com/sovakushwaha/ecommerce-transactions-eda

- Cleaned and analysed 6,000 transactions for revenue, returns, seasonality and performance patterns.
- Applied statistical tests (chi-square, Kruskal-Wallis, Mann-Whitney U, Spearman) to separate meaningful signals from descriptive noise.

2026

Purchase Behavior & RFM Segmentation (Python, SQLite)
github.com/sovakushwaha/purchase-behavior-analysis

- Processed 6,500 orders and built reusable Python modules for RFM scoring across 2,200 customers.
- Designed a structured analysis pipeline from data preparation through segmentation and quantitative interpretation.

2026

Field Service Operations Analytics (Python, SQLite, Streamlit)
github.com/sovakushwaha/field-service-operations-analytics

- Designed a normalised SQLite warehouse and analytical views for 3,200 work orders across technicians, categories and regions.
- Built interactive Streamlit reporting to communicate experimental / operational metrics clearly to stakeholders.

2026

EDUCATION

Master's Degree, Information Technology with Data Analytics
University of the West of Scotland Currently studying

Focus: data analytics, information technology, evidence-based decision-making and applied computing.

Jan 2026 to Present

Bachelor in Information Technology (BIT)
KIST College & SS Kathmandu, Nepal

Foundations: Programming, Algorithms and Data Structures, Database Systems, Mathematics, Software Development.

2021 to 2024

AWARDS & RECOGNITION

- Conference paper selected (CISCom 2026 oral; Springer CCIS): *An Explainable VGG6 Convolutional Neural Network for Multi-Class Ovarian Cyst Classification in Ultrasound Imaging.*
- 2nd Prize, KIST Inter-College IT Project Competition, SparkyBot AI Voice Assistant.